SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Comparative Analysis

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 3 – Comparative Analysis
Weightage:	30% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	CSE(AI)	**Date:**	19-08-26

Code of Conduct

I T. Kiran Kumar*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: T. Kiran Kumar

Instructions

- Read each scenario carefully before answering.
- Identify the NLP techniques being compared in the question.
- Analyze each approach based on its working principles, advantages, limitations, and applicability.
- Compare the alternatives using technical reasoning rather than listing definitions.
- Support your analysis with examples from the given scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze sentence representations, compare structural representations of language, and evaluate how different parsing approaches capture meaning and relationships among words.	Q1
Syntax-Driven Semantic Analysis	Analyze parsing strategies for incomplete and ambiguous inputs, evaluate parsing efficiency, and determine suitable methods for real-time language understanding.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze ambiguity in natural language sentences, evaluate ambiguity resolution techniques, and determine the most effective approach for selecting intended meanings.	Q3
Representing Linguistically Relevant Concepts	Analyze grammatical constraints, agreement features, argument structures, and linguistic representations required for accurate language processing.	Q4
Information Retrieval & Syntax-Driven Semantic Analysis	Analyze large-scale parsing strategies, evaluate performance trade-offs, and justify efficient approaches for processing massive linguistic datasets.	Q5

Annexure A – Comparative Analysis

1. A conversational AI system must identify grammatical relationships in user queries containing long-distance dependencies. The developers are comparing constituency parsing and dependency parsing.

Analyze how the two representations differ in capturing hierarchical and word-to-word relationships, and justify which approach is more suitable for extracting meaningful relationships from conversational text.

2. An NLP system processes partially typed search queries such as "best hotels near…" and "book a flight from Chennai to…". The system currently uses conventional top-down parsing, while the developers are considering Earley parsing.

Analyze how the two approaches behave when input is incomplete or ambiguous, and justify which parsing strategy is more appropriate for interactive NLP applications.

3. A grammar-checking system must distinguish between multiple interpretations of structurally ambiguous sentences. The developers are comparing CFG, PCFG, and neural constituency parsers.

Analyze how each approach represents and resolves syntactic ambiguity, and justify which approach would provide the best balance between interpretability and practical accuracy.

4. A multilingual NLP system must process sentences in which grammatical information depends on number, gender, tense, and person. The developers are considering feature structures and traditional context-free grammar rules.

Analyze how feature structures extend basic grammar representations and justify their usefulness for enforcing grammatical constraints in multilingual systems.

5. A voice assistant must interpret commands containing verbs with different argument requirements, such as "give the book to John," "sleep," and "put the file on the desk." The developers are considering subcategorization frames.

Analyze how subcategorization information helps determine valid sentence structures and justify its importance in semantic and syntactic analysis.

1. CONSTITUENCY PARSING VS DEPENDENCY PARSING

Constituency parsing represents a sentence as a hierarchical tree of phrases such as noun phrases (NP) and verb phrases (VP). It shows how words are grouped into larger grammatical units.

Dependency parsing represents a sentence using direct word-to-word relationships. Each word is connected to another word, usually with the main verb as the root.

For example, in the sentence “The student solved the problem”, dependency parsing directly shows that “student” is the subject of “solved” and “problem” is its object.

Constituency parsing is useful for understanding the overall hierarchical structure, while dependency parsing is more direct for identifying grammatical relationships between words.

For conversational AI, dependency parsing is generally more suitable for extracting meaningful relationships because it directly identifies subjects, objects, modifiers, and other grammatical relationships. It is also useful for information extraction and intent understanding.

2. TOP-DOWN PARSING VS EARLEY PARSING

Traditional top-down parsing starts from the grammar's start symbol and tries to generate a structure that matches the input. It can become inefficient when the input is incomplete or ambiguous because it may explore many possible grammar rules.

Earley parsing is a chart-parsing algorithm that can handle context-free grammars and is suitable for ambiguous and incomplete input. It stores partial parsing results and avoids repeating the same work.

For example, queries such as “best hotels near…” and “book a flight from Chennai to…” are incomplete. Earley parsing can maintain partial hypotheses and continue parsing when additional words are entered.

Therefore, Earley parsing is more appropriate for interactive NLP applications because it supports partial input, ambiguity, and incremental parsing more effectively than conventional top-down parsing.

3. CFG VS PCFG VS NEURAL CONSTITUENCY PARSING

A Context-Free Grammar (CFG) represents syntactic structures using grammar rules such as:

$S \rightarrow NP\ VP$

CFG can represent multiple valid structures for an ambiguous sentence but does not provide probabilities for choosing between them.

A Probabilistic Context-Free Grammar (PCFG) assigns probabilities to grammar rules. When multiple parses are possible, the parser can select the most probable structure.

A neural constituency parser uses machine-learning models and contextual representations to predict syntactic structures. It can learn complex patterns from large datasets and usually provides better practical accuracy.

Comparison:

CFG:
- Simple and interpretable
- Cannot rank different interpretations

PCFG:
- Can rank possible parses
- Depends on grammar probabilities

Neural Parser:
- High practical accuracy
- Less interpretable

For a grammar-checking system, a neural constituency parser provides the best practical accuracy, while CFG or PCFG rules can be used when explicit grammatical explanations are required.

4. FEATURE STRUCTURES VS TRADITIONAL CFG

Traditional CFG represents grammatical structures using rules such as:

S → NP VP

However, basic CFG does not naturally represent grammatical features such as number, gender, tense, and person.

Feature structures extend grammar rules by attaching attributes and values to words or phrases.

For example:

NP: [Number = Singular, Gender = Female]

VP: [Person = Third, Number = Singular]

The system can then enforce agreement between related elements.

For example:

“The boy plays.” is correct.

“The boy play.” is incorrect because the singular subject requires the appropriate verb form.

Feature structures are especially useful in multilingual NLP because languages may contain gender, number, case, person, and tense distinctions.

Therefore, feature structures provide a more expressive representation than basic CFG and are useful for enforcing grammatical constraints and agreement in multilingual systems.

5. SUBCATEGORIZATION FRAMES IN VOICE ASSISTANT COMMANDS

Subcategorization describes the arguments that a verb requires to form a valid sentence. Different verbs have different argument requirements.

“Give the book to John”

The verb “give” can require:
- Agent
- Object
- Recipient

Structure:

give + object + recipient

“Sleep”

The verb “sleep” does not require an object.

Structure:

sleep + subject

“Put the file on the desk”

The verb “put” requires:
- Agent
- Object
- Location

Structure:

put + object + location

Subcategorization helps the NLP system determine which sentence structures are grammatically valid and identify the roles of different arguments.

It is important for both syntactic and semantic analysis because it connects a verb with the types of participants required by the action.

Therefore, subcategorization frames are essential for voice assistants because they help correctly interpret commands and identify missing or incorrect arguments.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Scope of Items	Comparison Depth	Use of Evidence	Critical Thinking	Clarity of Presentation	Sub. Total	Total %
1	4	4	4	3	3	18	92
2	4	4	4	3	3	18	
3	4	4	4	3	3	19	
4	4	4	3	3	4	19	
5	4	4	3	3	4	18	

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